

Relational-Indexed Retrieval-Augmented Generation (RI-RAG):

Scalable Retrieval Efficiency and Memory Consumption Evaluation

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September 2026

1 Research Problem to Be Solved

Relational-Indexed Retrieval-Augmented Generation (RI-RAG) has already been developed as a working minimum viable prototype. Its architecture separates semantic retrieval from authoritative data storage: a vector database is used to identify relevant records, while the corresponding textual payload and relational metadata are stored in a relational database.

The next research phase is to determine whether this architecture provides measurable advantages at larger scales without introducing an unacceptable retrieval-time penalty.

The research will investigate two primary questions:

1. **Retrieval efficiency:** Can RI-RAG retrieve relevant information within a competitive time compared with conventional vector-database RAG?
2. **System-level memory consumption:** Does reducing the amount of payload stored in the vector database translate into a measurable reduction in the overall RAM footprint of the complete RAG system, rather than only reducing the memory associated with the vector-store representation?

The current prototype provides preliminary evidence that separating vector representations from textual payloads can reduce vector-storage memory. However, larger-scale experiments are necessary to determine whether this architectural property persists when the dataset grows from thousands to tens or hundreds of thousands of records.

The research will therefore evaluate RI-RAG against a conventional RAG architecture under controlled experimental conditions, focusing on retrieval latency, RAM consumption, storage requirements, retrieval accuracy, semantic-summary quality, and scalability.

2 Expected Resource Usage

2.1 Datasets

Two principal dataset scales are planned:

- **10,000 records:** initial large-scale validation and benchmarking.
- **100,000 records:** scalability evaluation at a substantially larger dataset size.

The datasets will be generated and refined using AI assistance. This is necessary because the RI-RAG experiment uses a strict database projection that must be consistently populated across a large number of records. Manually creating and validating tens of thousands of records would require substantial time and could introduce inconsistencies in the experimental data.

Each record will follow the RI-RAG relational projection:

Table 1: RI-RAG relational projection.

Field	Description
<code>file_id</code>	Unique identifier for the record.
<code>file_name</code>	Name associated with the record.
<code>content</code>	Full textual content stored in the relational database.
<code>semantic_summary</code>	Concise semantic representation of the content.
<code>grade</code>	Evaluation or quality score associated with the record.
<code>comments</code>	Additional evaluation comments associated with the record.

The AI-assisted dataset-generation pipeline will focus on generating records that conform to this schema while maintaining consistency between the textual content, semantic summary, grade, comments, and record identifiers.

2.2 Content–Summary Validation

A specific validation experiment will measure the semantic relationship between each record’s `content` and its `semantic_summary`.

Both fields will be converted into embeddings using the same embedding model. Their cosine similarity will then be calculated to estimate how closely the generated summary represents the original content.

For each dataset, the experiment will report:

- Mean cosine similarity.
- Median similarity.
- Standard deviation.
- Similarity distribution.
- Lower and upper percentile values.
- Percentage of records meeting the selected validation criterion.

For a record i , the semantic similarity can be expressed as:

$$S_i = \frac{E(c_i) \cdot E(s_i)}{\|E(c_i)\| \|E(s_i)\|}$$

where $E(c_i)$ represents the embedding of the original content and $E(s_i)$ represents the embedding of its semantic summary.

This experiment will help determine whether the automatically generated summaries maintain a consistent semantic relationship with their source content at both 10,000 and 100,000 records.

Semantic similarity will be treated as a validation signal rather than as proof of factual correctness. Where appropriate, a smaller sample of records may also be subjected to additional AI-assisted content-coverage or factual-consistency evaluation.

The generated data will also be validated for:

- Correct database schema compliance.
- Valid and unique `file_id` values.
- Presence of all required fields.
- Consistency between `content` and `semantic_summary`.
- Valid `grade` values.
- Appropriate `comments` associated with each record.
- Successful ingestion into both the relational and vector components of RI-RAG.

2.3 AI-Assisted Research

The project will use AI assistance primarily for dataset generation and refinement, coding assistance, database optimization, and experimental preparation.

An allocation of approximately **\$300** is requested for AI and LLM usage. This will support:

- AI-assisted dataset generation and refinement.
- Coding assistance during the optimization phase.
- Database and retrieval optimization.
- Content–summary semantic validation.
- Assistance with experimental preparation and analysis.

The exact token usage will depend on the generation and validation strategy and will be measured during the project rather than treated as a fixed requirement.

2.4 Human Research Assistance

The project is currently developed independently. At the optimization stage, having another researcher or technical assistant involved would help accelerate dataset preparation, testing, optimization, and validation.

If appropriate, part of the research stipend may be used to support human research assistance during the optimization and validation phase.

2.5 Researcher Stipend

A **12-month research stipend of \$500 per month** is requested, for a total of **\$6,000**.

This stipend will support my continued full-time work on RI-RAG, including:

- Development and optimization of the RI-RAG implementation.
- Dataset preparation and experimental validation.
- Benchmarking against conventional RAG systems.
- Memory and latency analysis.
- Investigation of the relationship between payload size and memory footprint.
- Analysis of the trade-off between memory usage and retrieval latency.
- Documentation and dissemination of the research.

3 Expected Costs

The proposed total budget is **\$6,300**.

Table 2: Proposed project budget.

Category	Allocation
AI assistance and LLM usage	\$300
12-month research stipend	\$6,000
Total	\$6,300

The requested funding is focused on the resources that are most directly needed at the current stage of the project rather than on high-performance computing infrastructure. The existing RI-RAG MVP, sample datasets, and experimental code are already available.

The remaining work primarily requires additional time, AI-assisted development and dataset refinement, and human assistance during the optimization and validation phase.

4 Experimental Methodology

The research will compare two systems.

4.1 Conventional RAG

A conventional architecture in which the vector database stores embeddings together with the associated retrieval payload.

4.2 RI-RAG

A hybrid architecture in which:

1. The vector database stores the information required for semantic retrieval and record identification.
2. Vector search identifies candidate record identifiers.
3. The corresponding records are retrieved from the relational database.
4. The relational database remains the authoritative source for the textual payload and structured metadata.

Both systems will be evaluated under equivalent experimental conditions.

4.3 Principal Measurements

The experiments will measure:

- Retrieval latency.
- Average latency.
- Median latency.
- P95/P99 latency where applicable.
- Peak RAM consumption.
- Baseline RAM consumption.
- Incremental RAM consumption attributable to the RAG system.
- Vector-storage footprint.
- Total database/storage footprint.
- Retrieval precision and recall.
- End-to-end retrieval correctness.
- Content–summary semantic similarity.
- Scaling behavior from 10k to 100k records.

Particular attention will be given to the distinction between vector-store memory reduction and total system RAM reduction.

A reduction in vector payload size does not automatically imply an equivalent reduction in overall RAM consumption because the database engine, operating system, indexes, caches, embedding model, and other components also consume memory.

This distinction is therefore a central research question.

5 Experimental Questions

5.1 RQ1 — Retrieval Performance

Can RI-RAG maintain retrieval latency that is competitive with conventional vector-based RAG as the dataset increases from 10,000 to 100,000 records?

5.2 RQ2 — Memory Consumption

Does the reduction of payload stored in the vector database produce a measurable reduction in the overall RAM consumption of the complete RAG system?

5.3 RQ3 — Scalability

How do retrieval latency, memory consumption, and storage requirements change as the number of records increases?

5.4 RQ4 — Semantic Summary Quality

Do the generated `semantic_summary` fields maintain a consistent semantic relationship with their corresponding `content` fields at both dataset scales?

5.5 RQ5 — Architectural Trade-offs

What trade-offs exist between memory consumption, retrieval latency, storage requirements, and the additional relational-database retrieval step introduced by RI-RAG?

6 Project Timeline and Key Milestones

6.1 Milestone 1 — Efficiency Improvement and Experimental Infrastructure

The existing RI-RAG implementation will be optimized using the researcher’s own computer together with AI-assisted development and analysis.

Activities will include:

- Profiling the current implementation.
- Improving database access patterns.
- Optimizing vector retrieval.
- Reducing unnecessary data movement.
- Improving memory measurement methodology.
- Establishing reproducible benchmark procedures.
- Validating the conventional RAG baseline.
- Establishing a common evaluation protocol.

Deliverable: reproducible benchmark infrastructure capable of comparing conventional RAG and RI-RAG under controlled conditions.

6.2 Milestone 2 — 10,000-Record Dataset

A validated dataset containing approximately **10,000 records** will be generated.

Activities will include:

- AI-assisted dataset generation.
- Schema validation.
- Relational consistency validation.
- Content–summary semantic validation.
- Embedding generation.
- Database ingestion.
- Initial scalability experiments.

Deliverable: validated 10k-record benchmark dataset and comparative experimental results.

6.3 Milestone 3 — 100,000-Record Dataset

The experiment will then scale to approximately **100,000 records**.

The same evaluation protocol will be applied to determine whether the behavior observed at 10,000 records persists at a substantially larger scale.

The content–summary relationship will also be evaluated at this scale to determine whether dataset-generation quality remains consistent.

Deliverable: validated 100k-record benchmark dataset and scalability results.

6.4 Milestone 4 — Comparative Analysis

The experimental results will be statistically analyzed to determine:

- Whether RI-RAG maintains competitive retrieval latency.
- Whether the vector-storage reduction persists with increasing dataset size.
- Whether the reduction translates into lower total RAM consumption.
- How relational database hydration affects end-to-end latency.
- Whether content–summary semantic similarity remains consistent at scale.
- At what dataset scale the architectural differences become measurable.
- What trade-offs exist between memory consumption and retrieval latency.

6.5 Milestone 5 — Research Publication

The results will be consolidated into a research paper documenting:

- The RI-RAG architecture.
- Experimental methodology.
- Dataset-generation methodology.
- Content–summary validation.
- Experimental setup.
- Resource measurements.
- Scalability results.
- Limitations.
- Reproducibility considerations.

7 Anticipated Outcomes

The primary anticipated outcome is a rigorous evaluation of whether RI-RAG can provide measurable resource and architectural benefits at larger database scales.

The project will investigate whether RI-RAG can simultaneously provide:

- Retrieval performance competitive with conventional vector-based RAG.
- Reduced vector-store memory requirements.
- Potential reductions in overall RAG system resource consumption.
- Improved traceability through an authoritative relational data layer.
- Explicit separation between semantic retrieval and authoritative data storage.
- Access to structured relational metadata during retrieval.
- A controlled methodology for evaluating automatically generated RAG datasets.

The research will **not assume that these benefits necessarily occur**. Instead, the experiments will determine their magnitude and identify the conditions under which they occur.

An important outcome will therefore also be a clear characterization of RI-RAG’s limitations and trade-offs.

For example, if relational database hydration introduces additional latency that offsets the memory benefits, this will be reported as an experimental finding. Similarly, if vector-store memory reduction does not translate into proportional total-RAM reduction, the research will quantify that difference.

8 Expected Scientific Contribution

The proposed research aims to contribute an experimentally validated architecture for separating semantic indexing from authoritative payload storage in RAG systems.

The contribution is particularly focused on the relationship between:

semantic retrieval $\rightarrow$ relational hydration $\rightarrow$ memory consumption $\rightarrow$ end-to-end retrieval performance.

The research will also provide a systematic approach for validating the relationship between source content and automatically generated semantic summaries.

Rather than evaluating only retrieval accuracy, the study will examine RAG as a systems problem involving:

- Memory.
- Storage.
- Retrieval latency.
- Semantic representation.
- Data traceability.
- Scalability.

The resulting benchmark will provide quantitative evidence regarding the scalability characteristics of RI-RAG and conventional RAG under controlled dataset sizes.

9 Publication Plans

The current RI-RAG MVP and its initial experimental results are already planned for submission to **arXiv** as a preliminary research contribution.

Following completion of the larger-scale experiments, the expanded research will be developed into a substantially more comprehensive manuscript for potential submission to a **Springer** venue.

The larger study will provide additional experimental evidence, including:

- 10,000-record evaluation.
- 100,000-record evaluation.
- Resource profiling.
- Memory analysis.
- Retrieval-latency analysis.
- Content–summary semantic validation.
- Scalability analysis.
- Statistical comparison between conventional RAG and RI-RAG.

10 Collaboration Opportunities

The project is **open to collaboration**.

Potential collaborators may contribute to:

- Dataset construction and validation.
- RAG benchmarking.
- Database and systems optimization.
- AI infrastructure.
- Statistical analysis.
- Reproducibility.
- AI systems research.
- Academic publication.

The project is particularly open to collaboration with researchers and engineers working on RAG systems, vector databases, relational databases, AI infrastructure, information retrieval, and efficient AI systems.

11 Budget Summary

Table 3: Project budget summary.

Item	Amount
AI assistance and LLM usage	\$300
12-month research stipend	\$6,000
Total	\$6,300

The central objective is to establish, through controlled experiments at **10,000 and 100,000 records**, whether RI-RAG can maintain competitive retrieval performance while reducing resource consumption and providing the traceability benefits of an authoritative relational data layer.